

Lab Project 1

Trustworthy-aware Human in the Loop

Trustworthy and Responsible Artificial Intelligence - 2026/2027

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1 Introduction

The Trustwortthy and Responsible Artificial Intelligence (TRAI) course comprises: (i) a theoretical-practical component, evaluated in a written exam , and (ii) a practical work, consisting of the development of two lab projects, each one on at least two topics covered by the course.

This document describes the Proposal for Project 1, and present guidelines for its Design and Development.

Project 1 relies on the human involvement, oversight, and supervision of AI systems, achieved by using human in or on the loop as a support for achieving the various typical requirements of the regulations (EU, UNESCO, etc.) for trustworthy and responsible AI. Note that, Project 2 relies specifically on the two important requirements of explainability (XAI) and reliability, so Project 1 does not need to involve these two requirements.

2 Project Proposal

Project1 should address a problem related to trustworthy and responsible AI systems, that is, systems designed to foster proactive, reliable cooperation with humans while respecting human values, maintaining user control, and promoting human well-being. As previously mentioned, it must relate to human–AI cooperation, operationalized through the involvement, oversight, and supervision of AI systems by humans. This is typically achieved by incorporating humans in or on the loop as part of the mechanisms that support compliance with regulatory frameworks for trustworthy and responsible AI (e.g., EU, UNESCO, OECD). Other columns that support Trustworthy AI can also be involved, such as Transparency, Safety, Security, Accountability, etc.

The project scenario should involve either a single-agent or a multi-agent system, where some agents may be humans or represent humans. These agents—whether autonomous AI systems or learning agents capable of classification and decision-making (as defined in the provided class code within the Practical Lab resources)—should be designed to act trustworthily. In particular, they should learn under ethical guidance, ensuring fairness, accountability, privacy preservation, and the promotion of human, societal, and even planetary well-being.

By aligning what we ask to label with how we want models to behave, active learning can serve not only as a method for efficient training but also as a mechanism for ethical model improvement. In human-in-the-loop (HITL) and, more specifically, in active learning systems, the annotation process itself constitutes a key dimension of trust.

Below are some guidelines and design ideas based on the available code and potential system extensions:

- Begin with the provided active learning notebooks. Start with *interactive_labeling_noModAL.ipynb*. It enables real human interaction. *active_learning_full_tutorial.ipynb* includes a comparison of different query strategies that can be used to select which instances to provide to the human to get labels. It simulates interactivity with humans using ground truth data, with digits dataset. Both follow a pool-based active learning approach, but the stream-based can be also used.

- In the current version of *interactive_labeling_noModAL.ipynb*, there is a single agent (the learner), represented by an instance of the Learner class, which employs the Least Confidence query strategy and

the $RandomForestClassifier$, but it can employ other query strategies (see $active_learning_full_tutorial.ipynb$) and other machine learning algorithms.

- In the Cooperative AI setting, agents can request assistance from humans or other agents when they detect uncertainty and need more information.
- The agent currently relies on information gain–based query strategies, primarily optimizing accuracy. One may redefine these strategies to balance performance, fairness, and ethical behavior, that is, you may extend or modify the query strategies so that the acquisition (labeling) of new instances promotes ethics, fairness, or accountability rather than just accuracy.
- Currently, a single human is allowed to interact with the agent and teaches it. One can extend this to multiple humans that can be automatically and carefully selected by their expertise, availability, ethical behaviour, etc., to label instances.
- Choose a specific application domain and find an appropriate dataset to replace the current one (the digits dataset).
- If you want to create a multi-agent system, define multiple learners with different expertise — each trained on distinct datasets or different subsets of the same dataset (see the part related with the committee query strategy in the tutorial file).
- In case there are no real human available for interacting with the active learning agent(s), human annotators can be modeled either as:
  - Standard ML models trained on different data distributions, or
  - LLM-based agents fine-tuned for specific expertise.
- You may combine the existing information-gain query strategies with ethically guided query mechanisms to achieve uncertainty-, bias-, and fairness-aware active learning.
- You may integrate annotation quality control strategies so that annotators are selected according to various features, so that they can play a role in the improvement of the ethical behaviour of the system.

Any computational tools and data sources may be employed, as long as their use is well-justified and aligned with the project's ethical objectives.

3 Design and Development

The proposed project will be developed throughout the PL classes. Students are expected to perform several experiments, produce a piece of software, and write a scientific paper describing the outcomes of the project.

3.1 Submission Instructions

The software developed and the paper must be made available on the official site of the course, Inforestudante, as well as on GitHub (alternatives are accepted if previously required).

3.2 Evaluation

Overall, this project (Project 1) is worth **25%** of the final grade of the TRAI course (note that the two projects of the course worth globally **50%**, i.e., (**10**/**20**). The grade is split into the following:

- Project (including paper, software);
- Defence.

Evaluation for this project will consider the following **criteria**:

- **Relevance** of the topic.
- **Significance** of the contributions.
- **Novelty/Originality** of the problem and approaches.
- **Quality/Technical Soundness** of the performed experiments, developed software, claims and conclusions.
- **Clarity** of the paper and of running instructions.

3.3 Important dates

The projects has the following deadlines, with results to be submitted to the course area in *Inforestudante*: **5th, November, 2026**

Oral defences are scheduled to: **6th, November, 2025**.

3.4 On Writing the Paper/Report

Here are some guidelines for writing the final paper/report.

The paper should be written in English or Portuguese, and have **at most 8 pages in a 2-column format**[1], excluding the pages needed for bibliographic references.

Although not rigid, you may structure your paper into the following parts or in equivalent ones (not necessarily in the same number):

1. Introduction (including motivation, problem description, goal and approach; may reuse part of the Project Proposal);
2. Related Work (may include part of the Project Proposal);
3. Materials and Methods - Data & Approach (an overview of the data used and of the approach adopted for tackling the problem);
4. Experimental methods (setup, quantitative and/or qualitative evaluation, and their discussion);
5. Conclusions (stressing the main achievements and main difficulties faced, and a reflection on the relation between the work to Trustworthy and Responsible AI topics) and Future Work (what could still be done);
6. Bibliographic References.

More details about what this parts are described in the following subsections.

3.4.1 Introduction

Please keep in mind that, when you write a report/paper, if the reader is not excited by the introduction, paper is lost.

The function of the Introduction is to:

- Establish the context of the work being reported. This is accomplished by discussing the relevant primary research literature (with citations) and summarizing our current understanding of the problem you are investigating;

[1]Formatted according to the IJCAI guidelines, https://www.ijcai.org/authors_kit

- State the purpose of the work in the form of the hypothesis, question, or problem you investigated; and, briefly explain your rationale and approach and, whenever possible, the possible outcomes your study can reveal.

The Introduction must answer the questions,"What was I studying? Why was it an important question? What did we know about it before I did this study? How will this study advance the state of the art?"

A recipe for the structure of successful introductions:

- paragraph 1: motivation: broadly, what is problem area, why important? Establish the context, background and/or importance of the topic. Establish the context by providing a brief and balanced review of the pertinent published literature that is available on the subject.

- paragraph 2: Narrow down: what is the problem you specifically consider? (this paragraph might be included in paragraph 1)

- paragraph 3: Explain how others solved it. Indicate a problem, controversy or gap of knowledge in those approaches/solutions proposed by others.

- paragraph 4: Explain shortly how you are going to solve it. This is the most crucial paragraph. Tell your elevator pitch, i.e., a summary that is short enough to give during an elevator ride: "In the paper, we...", "The purpose of this study was to..." or "We investigated three possible mechanisms to explain the ..."

- paragraph 5: how different/better/relates to other work. Provide a clear statement of the rationale for your approach to the problem studied. For example: State briefly how you approached the problem. Why did you choose this kind of experiment or experimental design? What are the scientific merits of this particular model system? What advantages does it confer in answering the particular question(s) you are posing? Do not discuss here the details. If you are using a novel (new, revolutionary, never used before) technique or methodology, the merits of the new technique/method versus the previously used methods should be presented here. (this paragraph might be included in the previous paragraph)

- paragraph 6: Provide an overview of the structure of the paper:"The remainder of this paper is structured as follows..."

3.4.2 Related Work

Discuss published work that relates to your project. How is your approach similar or different from others? It is of utmost importance that a submitted paper/report situates the work with respect to the current state of the art. Papers that are closely related to the presented work should be cited, discussed, and compared to the presented work. Although a superficial part of the related work is included in the Introduction, here you go deeper, with more details, more discussion, more comparison, etc.

3.4.3 Materials

Describe the data you are working with for your project. What type of data is it? Where did it come from? How much data are you working with? Did you have to do any preprocessing, or other special treatment to use this data in your project? Mention the tools that you use for implementing the project.

3.4.4 Methods

In the section Methods of a research article, you should give an account of how you carried out your research. The Methods should be clear and detailed enough for another experienced person to repeat the research and reproduce the results. As all of you implemented someway a system, this usually involves a description of:

- how you implemented it
- how you tested it (experimental setup and experimental methods).

This usually comprise two sections, the first titled, for instance, with the name of the system, and the second titled Experimental Setup or simply Experiments (in this case including a subsection with Experimental Setup).

A different/alternative scenario, and probably more common in AI, considers the subsection devoted to the experiments as a main section. In this case, the section Methods is confined to describe your approach for solving the problem that you set up in the introduction, while the section Experiments is devoted for describing the experiments that you performed to demonstrate that your approach solves the problem. The exact experiments will vary depending on the project, but you might compare your approach with previously published methods, perform an ablation study to determine the impact of various components of your system, experiment with different hyperparameters or architectural choices, use visualization techniques to gain insight into how your model works, discuss common failure modes of your model, etc. You should include graphs, tables, or other figures to illustrate

your experimental results. Note that this is still methodology, in this case, experimental methodology.

3.4.5 Results

Results should be clear and concise. The standard approach to this section is to merely present the results, without elaborate discussion or comment. This does not mean that you do not need any text to describe data presented in tables and figures. You may comment on the significant data presented in the tables and figures. This often takes the form of the location or summary statement, which identifies the table or figure and indicates its content. This is normally followed by a statement or statements which point out and describe the relevant or significant data. All your tables/figs should be numbered and given a title.

More elaborate commentary on the results is normally restricted to the Discussion section. In research articles, however, authors may comment extensively on their results as they are presented, and it is not uncommon for the Results section to be combined with the Discussion section under the heading: Results and Discussion

However, it is also possible that Results be the second of the two subsections of section Experiments, the first being Experimental Setup.

3.4.6 Discussion

This should explore the significance of the results of the work, not repeat them. A combined Results and Discussion section is sometimes appropriate.

3.4.7 Conclusions

The main conclusions of the study may be presented in a short Conclusions section, which may stand alone or form a subsection of a Discussion section, or of a Results and Discussion section. Conclusions are shorter sections of academic texts which usually serve two functions. The first is to summarise and bring together the main areas covered in the writing, which might be called "looking back"; and the second is to give a final comment or judgement on this. The final comment may also include making suggestions for improvement and speculating on future directions. Conclusions tend to include sections on significance of the findings and recommendations for future work. It is therefore valuable to honestly identify the limitations of your work and possible directions to solve this limitations.

3.4.8 Aknowledgements

You should include a statement on the use of artificial intelligence tools such as: "This document was slightly refined with the assistance of Large Language Models, such as ChatGPT, or similar tools (Grammarly), which helped check grammar, correct typos, and enhance clarity. Any Artificial Intelligence-generated text or content is clearly indicated within the document and is used only to a limited extent. The overall content and ideas remain solely the responsibility of the author."